

Starting the Game Project

We'll begin building the game by creating an empty Pygame window. Later, we'll draw the game elements, such as the ship and the aliens, on this window. We'll also make our game respond to user input, set the background color, and load a ship image.

Creating a Pygame Window and Responding to User Input

We'll make an empty Pygame window by creating a class to represent the game. In your text editor, create a new file and save it as *alien_invasion.py*; then enter the following:

```
alien
_invasion.py

import sys

import pygame

class AlienInvasion:
    """Overall class to manage game assets and behavior."""

    def __init__(self):
        """Initialize the game, and create game resources."""
        ❶ pygame.init()

        ❷ self.screen = pygame.display.set_mode((1200, 800))
        pygame.display.set_caption("Alien Invasion")

    def run_game(self):
        """Start the main loop for the game."""
        ❸ while True:
            # Watch for keyboard and mouse events.
            ❹ for event in pygame.event.get():
                ❺ if event.type == pygame.QUIT:
                    sys.exit()

            # Make the most recently drawn screen visible.
            ❻ pygame.display.flip()

if __name__ == '__main__':
    # Make a game instance, and run the game.
    ai = AlienInvasion()
    ai.run_game()
```

First, we import the `sys` and `pygame` modules. The `pygame` module contains the functionality we need to make a game. We'll use tools in the `sys` module to exit the game when the player quits.

Alien Invasion starts as a class called `AlienInvasion`. In the `__init__()` method, the `pygame.init()` function initializes the background settings that Pygame needs to work properly ❶. Then we call `pygame.display.set_mode()` to create a display window ❷, on which we'll draw all the game's graphical elements. The argument `(1200, 800)` is a tuple that defines the dimensions of the game window, which will be 1,200 pixels wide by 800 pixels high. (You can adjust these values depending on your display size.) We assign this